

WASP-10b

R-1728

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_130921 · created 2026-08-02T17:23:34+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456556.7913 +/- 0.002 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.209 +/- 0.06
Transit depth (Rp/Rs)^2	4.37 +/- 2.5 [%]
Orbital Inclination (inc)	89.38 +/- 1.84
Airmass coefficient 1 (a1)	1.05 +/- 0.047
Airmass coefficient 2 (a2)	-0.05 +/- 0.36
Scatter in the residuals of the lightcurve fit is	4.49 %
Best Comparison Star	#2 - [375, 135]
Optimal Aperture	2.15
Optimal Annulus	9.24
Transit Duration (day)	0.094 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.209 ± 0.06 vs 0.1592 (deviation 0.83σ)
Duration fit vs published	0.094 d vs 0.0946 d (deviation 0.05σ)
Mid-time O–C	5.1 min
Depth SNR	3.5
Residual scatter	4.49 %

Light curve

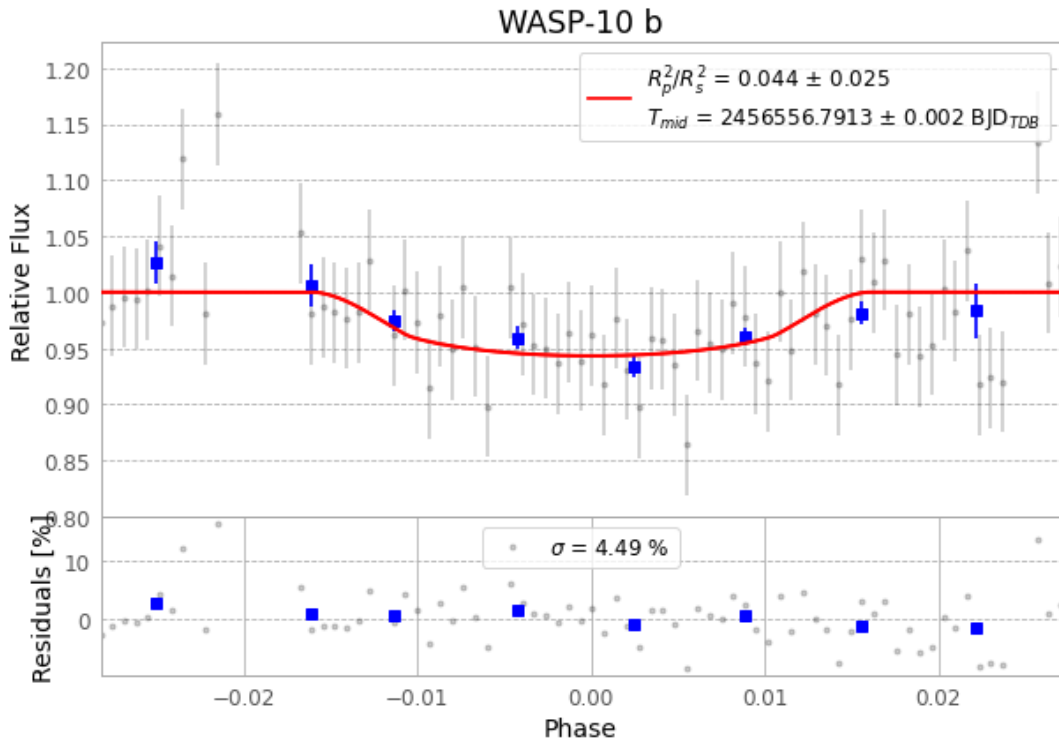


Figure 1 — FinalLightCurve_WASP-10 b_2013-09-20.png (SHA-256 b2500bdb56b42c8b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [379, 233], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9debcd73fdecfb41...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

83 FITS frames (55 MB), WASP-10130921044512.FITS → WASP-10130921085112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-130921.log (31085cfde78d...), FinalLightCurve_WASP-10 b_2013-09-20.png (b2500bdb56b4...), FinalLightCurve_WASP-10 b_2013-09-20.csv (7e105bfaa168...)

Compute resources

Wall time 360.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 17:23Z.